

Diploma in Computer Science & Information Technology

SBTE, Bihar — Semester IV

Environmental Education and Sustainable Development

Course Code: 2400006

Unit-wise Detailed Notes

UNIT 1: Ecosystem

1.1 Aquatic and Terrestrial Ecosystems

An ecosystem is a functional unit of nature consisting of living organisms (biotic) and their non-living environment (abiotic) interacting as a system.

Aquatic Ecosystem

Found in water bodies. Types: Freshwater (rivers, lakes, ponds, wetlands) and Marine (oceans, coral reefs, estuaries). Characteristics: temperature, dissolved oxygen, salinity, depth, light penetration determine the organisms present.

Terrestrial Ecosystem

Found on land. Types: Forest, Grassland, Desert, Tundra, Mountain. Characteristics: rainfall, temperature, and soil type determine the biome. Forests are the most productive and biodiversity-rich terrestrial ecosystems.

1.2 Structure of Ecosystem

- **Producers (Autotrophs):** Plants, algae — convert sunlight to chemical energy (photosynthesis).
- **Consumers (Heterotrophs):** Primary consumers (herbivores), Secondary consumers (carnivores), Tertiary consumers (top predators).
- **Decomposers:** Bacteria and fungi — break down dead organic matter and recycle nutrients.
- **Abiotic Components:** Sunlight, water, air, soil, temperature, minerals.

1.3 Food Chain and Food Web

Food Chain: Linear sequence of organisms where each is eaten by the next. Example: Grass → Grasshopper → Frog → Snake → Eagle. Energy flows in one direction; approximately 10% of energy transfers to the next level (10% Law).

Food Web: Interconnected network of food chains in an ecosystem. More realistic representation of feeding relationships. Each organism may have multiple predators and prey.

1.4 Biogeochemical Cycles

Carbon Cycle

Carbon moves through: Photosynthesis (CO_2 absorbed by plants), Respiration (CO_2 released), Decomposition, Combustion of fossil fuels (major source of CO_2 increase). Oceans are major carbon sinks.

Nitrogen Cycle

Nitrogen (N_2 , 78% of atmosphere) cannot be used directly by most organisms. Nitrogen fixation (bacteria convert N_2 to NH_3) → Nitrification (NH_3 to NO_3^-) → Plant uptake → Denitrification (back to N_2).

Sulphur Cycle

Sulphur released from rocks, volcanic activity, decomposition → SO_2 in atmosphere → acid rain → absorbed by plants. Burning fossil fuels releases SO_2 , causing acid rain.

1.5 Global Warming — Causes and Effects

Global warming is the gradual increase in Earth's average surface temperature due to the greenhouse effect enhanced by human activities.

Causes

- Burning fossil fuels (CO₂ release) — largest contributor.
- Deforestation reduces carbon absorption.
- Agriculture (methane from livestock, rice paddies).
- Industrial processes releasing greenhouse gases.

Effects

- Rising sea levels (melting ice caps) threatening coastal areas.
 - More frequent and intense extreme weather events.
 - Disruption of agriculture and food security.
 - Species extinction due to habitat loss and temperature change.
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UNIT 2: Air and Water Pollution

2.1 Water Pollution

Sources of Water Pollution

- Industrial effluents: Heavy metals (mercury, lead), chemicals, heat.
- Agricultural runoff: Pesticides, fertilizers causing eutrophication.
- Sewage and domestic waste: Pathogens, BOD increase.
- Oil spills in marine environments.
- Mining runoff: Acid mine drainage.

Effects of Water Pollution

- Waterborne diseases: Cholera, typhoid, dysentery.
- Eutrophication: Excessive algae growth depletes oxygen, killing aquatic life.
- Bioaccumulation of toxins in the food chain.

Water Quality Standards (IS)

Indian Standards for drinking water: pH 6.5-8.5, Turbidity <5 NTU, Total Dissolved Solids <500 mg/L, Total Hardness <300 mg/L, No coliform bacteria.

Water Pollution Control

- Sewage Treatment Plants (STP): Primary (screening, settling), Secondary (biological treatment), Tertiary (advanced filtration, disinfection).
- Effluent treatment before industrial discharge.
- Reducing agricultural chemical use.

2.2 Air Pollution

Sources of Air Pollution

- Vehicular emissions: CO, NOx, particulate matter, hydrocarbons.
- Industrial emissions: SO₂, NOx, heavy metals, dust.
- Combustion of fossil fuels in power plants.
- Agricultural activities: Ammonia, methane, burning of crop residues.
- Construction dust and mining.

Effects on Human, Plant, and Animal

- Humans: Respiratory diseases (asthma, bronchitis, lung cancer), cardiovascular disease, eye irritation.
- Plants: Reduced photosynthesis, leaf damage, crop yield reduction.
- Animals: Respiratory problems, reproductive failures.

Air Pollution Control

- Scrubbers: Remove SO₂ and particulates from industrial emissions.
 - Catalytic converters: Reduce vehicle exhaust pollutants.
 - Electrostatic precipitators: Remove particulates using electric charge.
 - Switching to cleaner fuels and renewable energy.
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UNIT 3: Sustainability and Renewable Sources of Energy

3.1 Concept of Sustainable Development

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs (Brundtland Commission, 1987). It balances economic growth, social equity, and environmental protection — the three pillars of sustainability.

3.2 Solar Energy

Solar energy is radiant energy from the sun. India receives abundant solar radiation — particularly favorable for solar power.

- **Photovoltaic (PV) Systems:** Solar cells convert sunlight directly to electricity using the photovoltaic effect. Used in solar panels for homes, commercial buildings, and utilities.
- **Solar Thermal Systems:** Use sunlight to heat a fluid. Applications: Solar water heater, solar dryer, solar cooker, solar still (desalination), solar pond.

3.3 Wind Energy

Wind energy converts kinetic energy of wind into electricity using wind turbines. India is among the top 5 wind energy producers globally. Major wind farms in Tamil Nadu, Gujarat, Rajasthan, Maharashtra.

- Advantages: Clean, renewable, no water consumption, can coexist with agriculture.
- Challenges: Intermittent (depends on wind speed), noise, bird/bat strikes, visual impact.

3.4 Biomass Energy

Biomass is organic material from plants and animals. Biomass energy is derived from this material through combustion, biogas production, or biofuel production.

- **Biogas:** Produced by anaerobic digestion of organic waste (cow dung, food waste). Main component: methane (60-70%). Clean cooking fuel for rural households.
- **Types:** Agricultural residues, wood, animal waste, municipal solid waste, energy crops.

3.5 Hydroponics and Aquaponics

Hydroponics: Growing plants without soil, using mineral nutrient solutions in water. Highly water-efficient (uses 70-90% less water than conventional agriculture). Aquaponics: Integrated system combining fish farming (aquaculture) with hydroponics. Fish waste provides nutrients for plants; plants filter water for fish.

3.6 New Energy Sources

- **Hydrogen Energy:** Hydrogen burned as fuel produces only water. Challenges: storage and production efficiency.
 - **Ocean Energy:** Tidal energy (tides driving turbines), Wave energy, Ocean Thermal Energy Conversion (OTEC).
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UNIT 4: Climate Change and Sustainable Development

4.1 Impact of Climate Change

- Rising sea levels threatening coastal cities and island nations.
- More frequent and severe droughts, floods, hurricanes.
- Shifts in agricultural zones affecting food security.
- Loss of biodiversity and ecosystem disruption.
- Increased spread of vector-borne diseases.
- Melting of Arctic/Antarctic ice and glaciers affecting freshwater supplies.

4.2 Factors Contributing to Climate Change

Greenhouse Effect: Solar radiation warms the Earth; GHGs (CO₂, CH₄, N₂O, water vapor) trap outgoing infrared radiation. Key greenhouse gases: Carbon dioxide (CO₂) — fossil fuel combustion, Methane (CH₄) — agriculture and livestock, Nitrous oxide (N₂O) — fertilizers, Fluorinated gases — industrial processes.

4.3 Sustainable Development Goals (SDGs)

The UN's 17 SDGs adopted in 2015 (Agenda 2030) provide a framework for global sustainable development. Relevant SDGs include: SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities), SDG 13 (Climate Action), SDG 14 (Life Below Water), SDG 15 (Life on Land).

4.4 India's Action Plan on Climate Change (NAPCC)

India launched the National Action Plan on Climate Change (NAPCC) in 2008 with 8 national missions: Solar Mission, Enhanced Energy Efficiency, Sustainable Habitat, Water Mission, Himalayan Ecosystem, Green India, Sustainable Agriculture, Strategic Knowledge for Climate Change.

UNIT 5: Environmental Legislation and Sustainable Building Practices

5.1 Environment Management System (EMS)

EMS (ISO 14001) is a structured framework for organizations to manage environmental responsibilities. Key elements: Environmental policy, Planning (objectives, targets), Implementation, Monitoring and measurement, Management review. EMS helps reduce environmental impact and improve regulatory compliance.

5.2 Green Building Concept

A green building is designed, constructed, and operated to minimize environmental impact and resource consumption while maximizing occupant comfort and health. Green building rating systems: LEED (Leadership in Energy and Environmental Design), GRIHA (Green Rating for Integrated Habitat Assessment — India's own system).

Key features: Energy efficiency, Water efficiency, Indoor air quality, Use of sustainable materials, Site selection and planning.

5.3 Green and Sustainable Building Materials — 5R Concept

- **Reduce:** Minimize resource consumption and waste generation during construction.
- **Reuse:** Use reclaimed materials (bricks, timber, steel) from demolished structures.
- **Recycle:** Use materials made from recycled content (fly ash bricks, recycled steel).
- **Refuse:** Avoid harmful and non-sustainable materials.
- **Rethink:** Innovate and redesign for greater sustainability and efficiency.

5.4 Environmental Protection Laws in India

- **Environment Protection Act, 1986:** Umbrella legislation for environmental protection. Central government can set standards, regulate activities, and take legal action.
- **Water (Prevention and Control of Pollution) Act, 1974:** Establishes Central and State Pollution Control Boards for water quality.
- **Air (Prevention and Control of Pollution) Act, 1981:** Regulates air pollution from industries and vehicles.
- **Wildlife Protection Act, 1972:** Protects wildlife, prohibits hunting of endangered species.
- **Forest Conservation Act, 1980:** Prevents diversion of forest land for non-forest purposes.

5.5 Carbon Footprint and Carbon Credit

Carbon Footprint: Total greenhouse gas emissions (in CO₂ equivalent) caused by an individual, organization, or product. Zero Carbon Footprint Building: Produces as much energy as it consumes from renewable sources; net zero emissions.

Carbon Credit: A tradeable permit representing the right to emit one tonne of CO₂. Organizations that reduce emissions below their allowance earn credits they can sell. Creates economic incentive for emission reduction. Part of carbon markets under the Kyoto Protocol and Paris Agreement.
